

Database Design Project

Oracle Baseball League Store Database

Project Scenario:

You are a small consulting company specializing in database development. You have just been awarded the contract to develop a data model for a database application system for a small retail store called Oracle Baseball League (OBL).

The Oracle Baseball League store serves the entire surrounding community selling baseball kit. The OBL has two types of customer, there are individuals who purchase items like balls, cleats, gloves, shirts, screen printed t-shirts, and shorts. Additionally customers can represent a team when they purchase uniforms and equipment on behalf of the team.

Teams and individual customers are free to purchase any item from the inventory list, but teams get a discount on the list price depending on the number of players. When a customer places an order we record the order items for that order in our database.

OBL has a team of three sales representatives that officially only call on teams but have been known to handle individual customer complaints.

Section 3 Lesson 3 Exercise 4: Normalization and Business Rules

Using Normalization to Validate Data (S3L3 Objective 3)

Third Normal Form

Normalization is a relational database concept, but its principles apply to data modeling.

- Third Normal Form (3NF)
 - It met all database requirements for both 1NF and 2NF.
 - No non-UID attribute can be dependent on another non-UID attribute.
 - Each column must depend directly on the UID. All attributes that are not dependent on the UID must be removed. For example, attributes that can be derived from data contained in other fields and tables must be eliminated. (All transitive dependencies are eliminated).

Take the data that is currently represented in the ERD in 2NF and apply the principles of Third Normal Form so that the data can be said to be in 3NF.